

HEALTHCARE AI & COMPUTER VISION

Project Overview

Deep-learning medical imaging application analyzing uploaded dermatological images to classify multi-class skin conditions using a Convolutional Neural Network.

Problem Statement:	Dermatological consultations are often inaccessible in rural areas, delaying early identification of treatable conditions.
Solution Implemented:	Built a computer vision web pipeline with OpenCV preprocessing, TensorFlow CNN model inference, and detailed confidence score outputs.
My Technical Role:	AI / ML Developer & Model Specialist

Main Features & Technical Stack

Core Features:	Technologies Used:
- Image drag-and-drop uploader with client preview - OpenCV image normalization and pre-processing pipeline - Multi-class CNN model evaluation with confidence breakdown - Educational dermatological reference & medical disclaimer	Python, FastAPI, TensorFlow, Keras, OpenCV, CNN Architecture, Render
	Architecture Flow:
	Web Client → FastAPI Image Ingestion → OpenCV Preprocessor → TensorFlow CNN → Softmax Confidence Evaluator

Technical Challenge & Results

Key Engineering Challenge: Quantized model weights to TFLite format to reduce memory footprint by 65% for free-tier cloud deployment.
Validated Results: Lightweight computer vision model returning skin condition predictions in under 1.2 seconds.

Live Demo: <https://github.com/prashant-singh-78/skin-desease.ml.git> | **GitHub:** <https://github.com/prashant-singh-78/skin-desease.ml.git> | **Portfolio:** prashant08.vercel.app